FOUNDERRADAR

Decoding the DNA of Successful Startup Teams

Group 12:

Nguyen Anh Duc
Ngo Dinh Khanh
Nguyen Dai Nghia
Lai Duc Minh

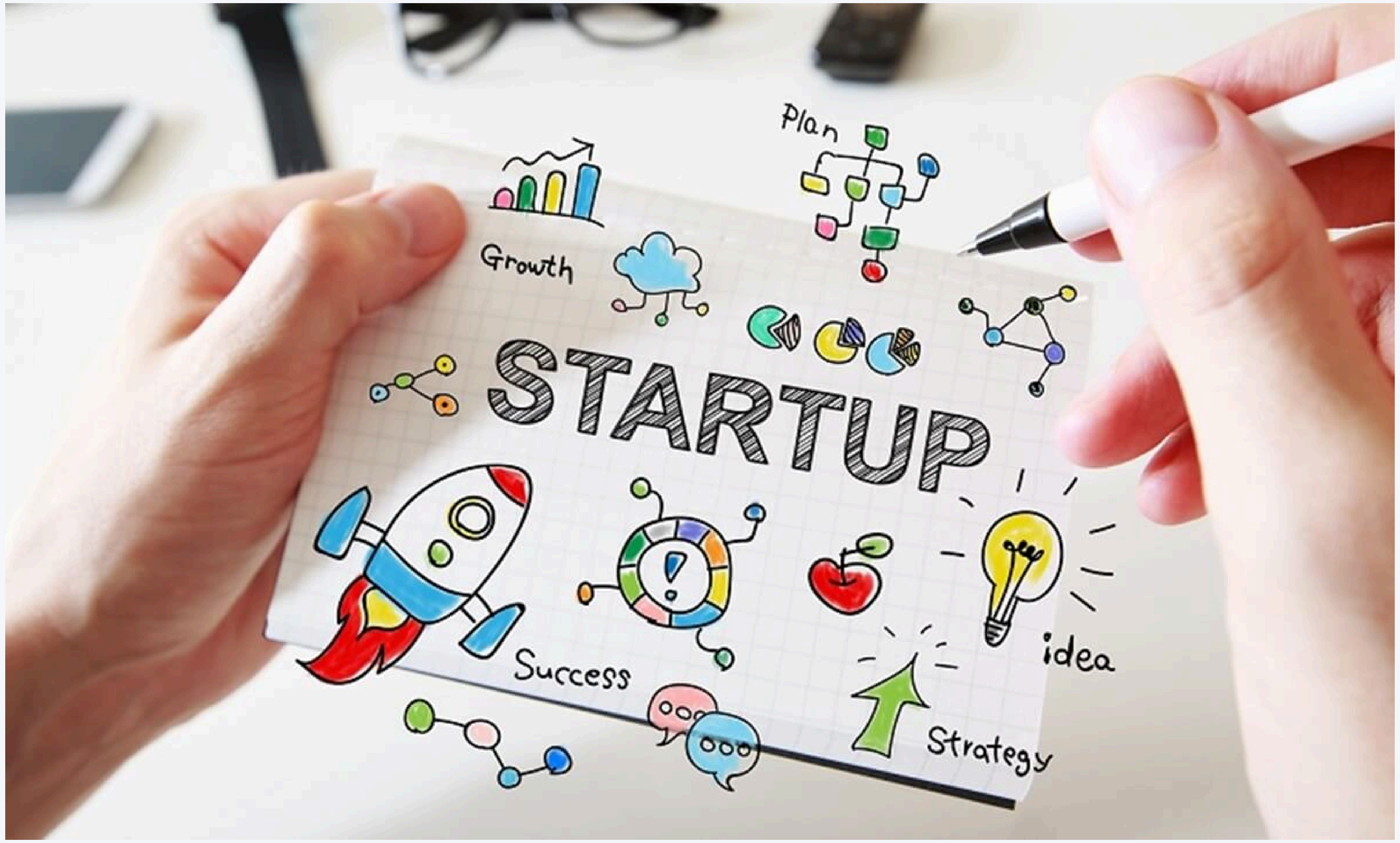
GitHub:
https://github.com/khanhngoo/FounderRadar

Problem & Product Vision

Motivation: Why This Project?

Most startup dashboards show funding and company metadata, but they rarely connect founder background, team signals, sector momentum, geography, and exit outcomes in one view.

That’s why we built **FounderRadar**!! We want to help users explore the human-capital patterns behind startup success, not to give a final investment answer, but to support faster discovery and deeper questioning.



Data Foundation

Dataset: What We Analyze

We use two complementary data layers:

- The **YC Ecosystem** layer contains 6,389 YC-backed companies and supports opportunity discovery through cohort, sector, region, hiring, and radar-score signals.

- Y Combinator Startup Directory (2005 - 2026)
- 2024 YCombinator All Companies Dataset
- Y Combinator Jobs Enriched

- The **Startup Intelligence** layer contains 42,751 modeled startup records and 3,250 success exits, built from company metadata, funding rounds, acquisitions, IPOs, founder relationships, and degree records.

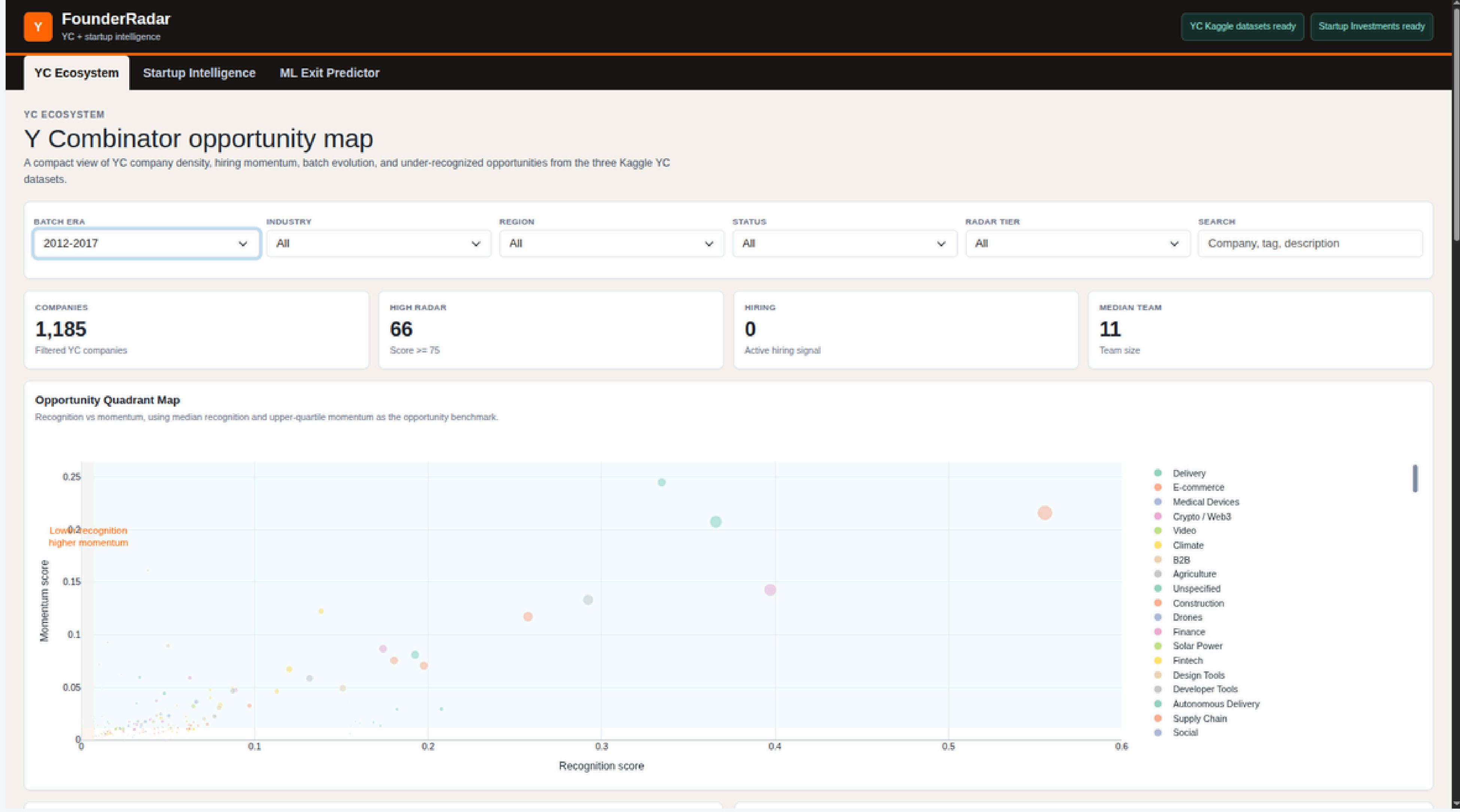
- Startup Investments kaggle - justinas

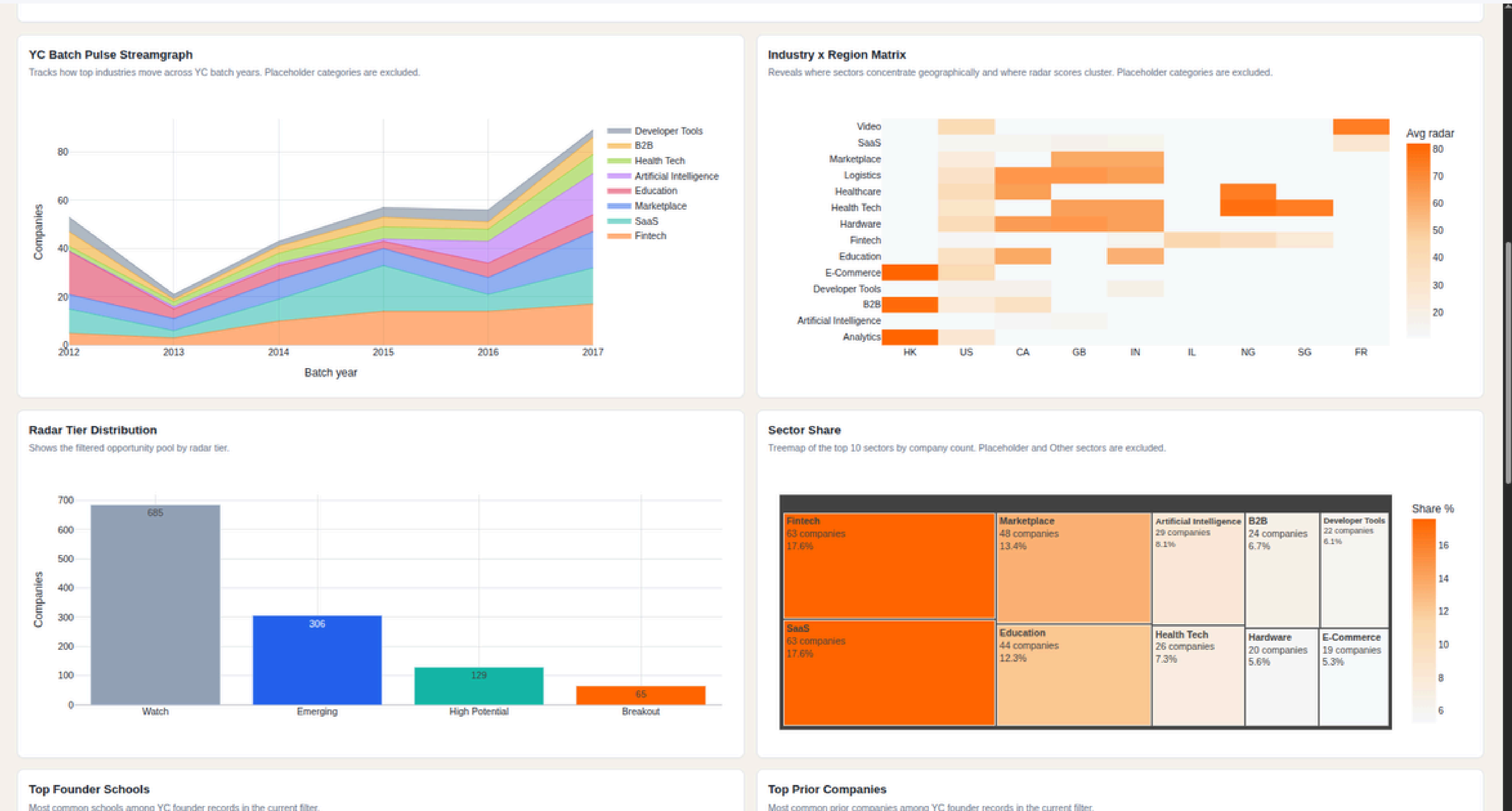
YC Ecosystem Dashboard

Charts We Use and Why

For the YC Ecosystem, we use an **opportunity quadrant map** to separate recognition from momentum, so under-recognized but fast-moving companies stand out.

We also use **streamgraphs** to show how YC sectors change across batches, **heatmaps** for industry-region concentration, **treemaps** for sector share, and **bar charts** for radar-tier distribution.





Startup Intelligence Dashboard

Charts for Historical Startup Signals

Exit Proxy Model

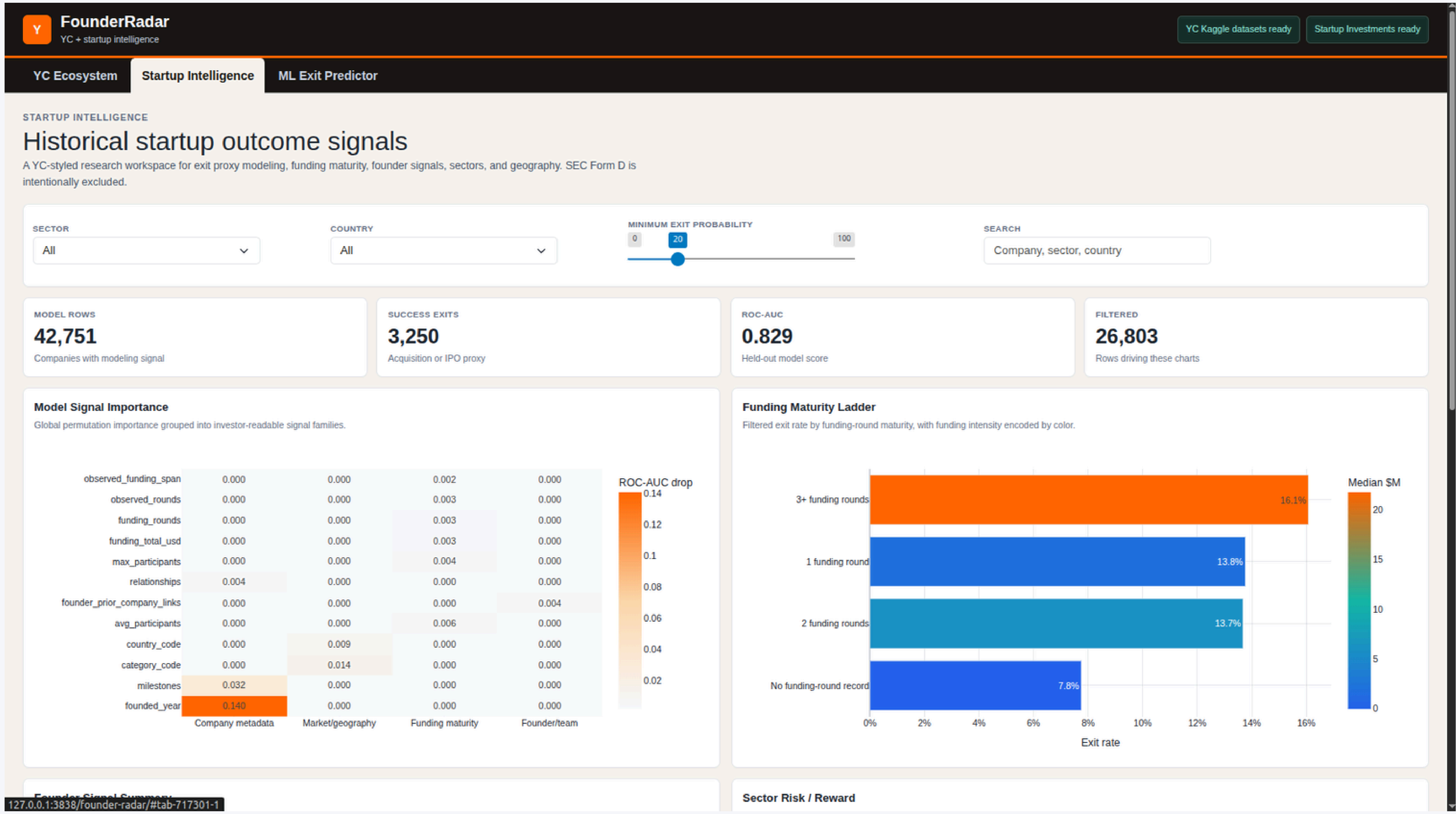
In Startup Intelligence, **heatmaps** explain which feature groups affect the exit model, **horizontal bars** show how exit rate changes with funding maturity, and **bubble charts** compare sector scale, exit rate, and funding. These choices help users compare patterns without overloading them with raw tables first.

From Chart to Decision

The goal is not only to show charts, but to guide an analysis workflow: overview first, filter next, compare patterns, then inspect company-level records.

Why These Encodings?

We use **position** for precise comparison, **size** for volume, and **color** for categories or intensity. For example, larger bubbles make high-volume sectors visible, while color gradients make funding or radar-score differences easier to scan.



Interactivity & Accessibility

Interactive Filtering

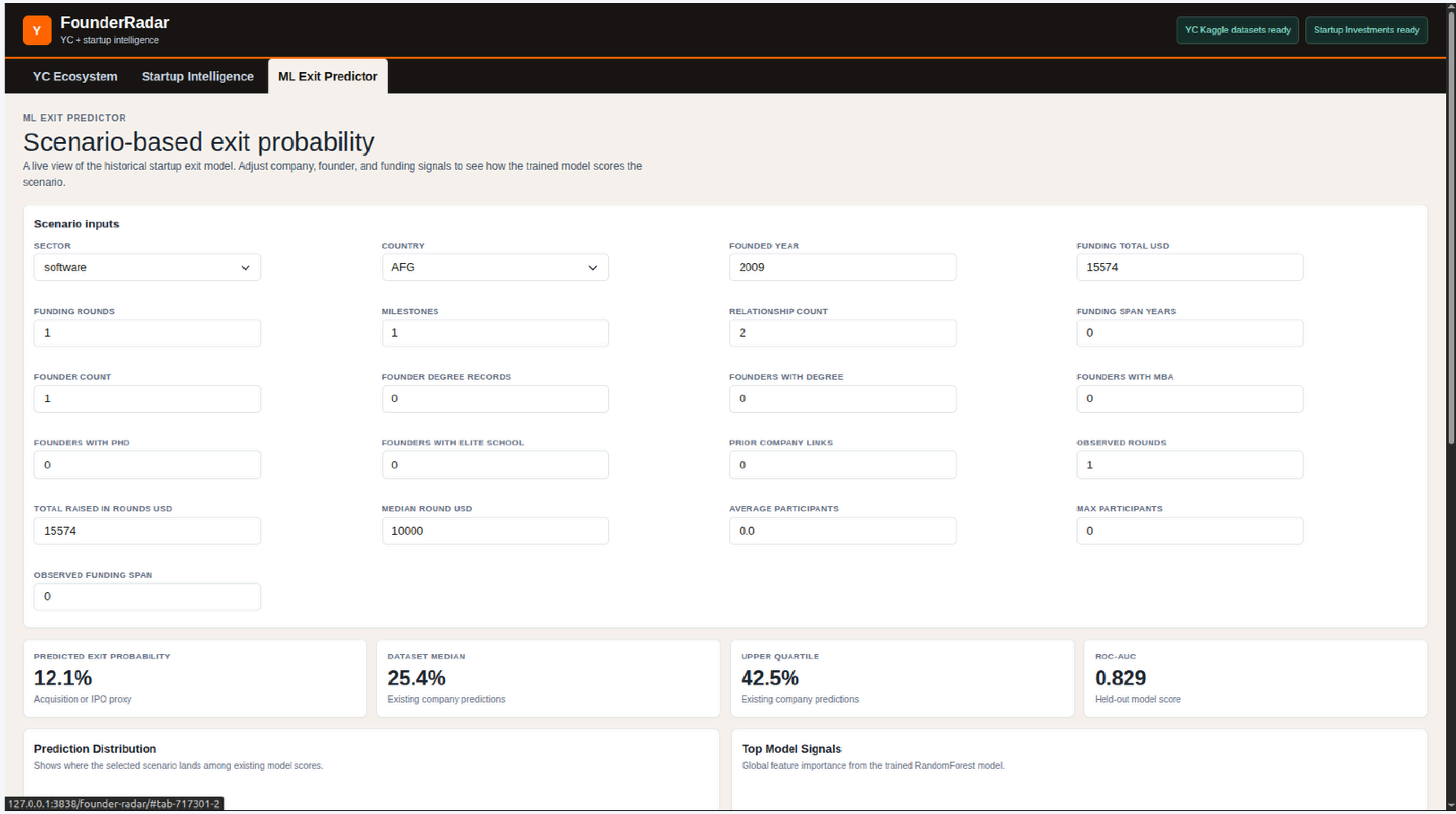
Users can filter by batch era, industry, region, status, radar tier, sector, country, and minimum exit probability. This lets users narrow the dataset while the KPI cards, charts, and tables update around the same analytical question.

Accessibility Choices

We maintain strong text contrast, use readable sans-serif typography, and avoid relying only on color. Labels, tooltips, chart position, and size help users interpret information even when colors are hard to distinguish.

Predictive Analytics

A Random Forest Classifier analyzes founder background signals to estimate the probability of startup success based on historical data patterns.



Technical Rigor: Briefly

Stack

Built with Python Shiny, Pandas, NumPy, Plotly, and scikit-learn.

Model

A balanced RandomForestClassifier estimates historical exit probability from funding, sector, geography, and founder/team signals.

Performance

Expensive joins, projections, and model steps are cached or computed at startup to keep the dashboard responsive.

Design Principle

We keep the interface data-dense but structured: filters first, KPI overview, chart comparison, then detailed records.

Limitation

The radar score and exit probability are analytical signals, not investment recommendations.

Why It Matters

This technical layer supports trust, speed, and auditability, but we only need to present it briefly before moving to the demo.

Findings & Lesson Learnt

Why It Matters

Findings: FounderRadar works best as an exploratory tool: combining founder, funding, sector, geography, and outcome signals reveals patterns that single charts or raw tables cannot show clearly.

Lessons learned: Good data visualization is not only about attractive charts. It requires a clear analytical workflow, careful data preparation, responsible model explanation, and accessible interaction design.

Thank you for listening!!